UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the October/November 2008 question paper

5090 BIOLOGY

5090/06 Paper 6 (Alternative to Practical), maximum raw mark 40

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1 **(a)** by watering;
by covering seedling/leaves (with a bag etc.);
A: vaseline/shade/darkness. R: cold/negatives. [2]

(b) **(i)** water absorbed until no more available;
transpiration/evaporation/surface loss (continues);
seedling/cells lose water/become flaccid; [up to 2]

(ii) root hairs damaged/broken;
so that no water absorbed; [2]
root hairs once in **(ii)** or **(iii)**, otherwise A: root

(iii) new root hairs grow (overnight);
R: ref conditions. [1]

(iv) fertiliser solution reduced water potential (in soil);
making endosmosis impossible (AW); [2]

(v) give seedling/surrounding soil,
more water/raising external water potential;
R: if fertiliser mentioned. [1]

(c) Two key points needed:
range of concentrations (of fertiliser solutions) prepared;
at least 3 days for results to show;

With up to three more marks from:
seedlings grown in pots/other containers;
name of plant/fertiliser used;
uniform size;
uniform condition(s), etc.;
how fertiliser applied/used for watering, etc.;
replication;
result recording/tabulation; [up to 5]

[Total: 15]

2 Colour of contents:

(i) **A1 (a)** blue; **A1 (b)** blue; **A2** (blue)-black; [3]
For **A2** accept iodine colour, brown, orange

(ii) Table 2.1 – conclusions
A1 (a): reducing sugar/glucose present;
A1 (b): no protein/protein absent;
A2: starch present; [3]

(iii) Table 2.2 – conclusions
B1 (a): no reducing sugar/glucose;
B1 (b): protein present;
B2: fat (oil)/lipid present; [3]

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(iv) chopping increases surface area/releases cell contents;
dissolved in ethanol;
water added to complete test;
if water added to original, ethanol goes milky/AW; [up to 2]

[Total: 11]

3 (a) (i) blood; [1]

(ii) leucocyte/wbc/polymorph/granulocyte/phagocyte/lymphocyte;
erythrocyte/rbc;
platelet;
plasma; R: serum, etc. [4]

(iii) Drawing marks:
1. at least 5 cm, realistic, clear and clean lines.
2. accurate nucleus – 3 main lobes. [D.2]

Labels: nucleus;
cytoplasm/cell membrane; R: if cell wall shown [2]

(iv) both dimensions accurate, with units at least once;
R: if no indication on drawing where measured.
[10 mm ± 1 mm/1.0 cm ± 0.1 cm on Fig. 3.1]
correct expression – drawing over image ; [e.g. 50/10]
allowance for × 750;
correctly stated mag; [750 × 5 =] × 5 / × 3750
[must be whole number for 3000 ÷,
up to 2 d.p. and 0.2 rounding if no allowance for × 750] [4]

(b) (i) transport of oxygen; [1]

[Total: 14]

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